Shuai Shao

Ph.D. Candidate / Researcher in Data Science · UNC Charlotte

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RESEARCH INTERESTS

AI in Mental Health: chatbot safety, interaction fine-tuning, transference, and LLM agents for suicidal ideation detection.
AI literacy: metaphorical framing of AI use and AI discourse on social media.
AI for Construction Safety: vision-language models, construction-site video analysis, and comparative model evaluation.
AI for Metropolitan Transportation: city data assessment, transportation modeling frameworks, and stakeholder engagement.

EDUCATION

University of North Carolina at Charlotte

Ph.D. in Data Science · 2025-present

M.S. in Data Science and Business Analytics · 2024-2025

Xiamen University

B.A. in French Language and Culture; Economics minor · 2007-2011

RESEARCH EXPERIENCE

AI in Mental Health · Planned Dissertation Focus

December 2024-present

Ongoing research on safe and responsible mental health support chatbots, human-AI interaction, and LLM agents for suicidal ideation detection. This line of work is the planned focus of my dissertation.

DragSI · LLM Agents for Suicidal Ideation Detection

First-author research · Manuscript under review

Designed and implemented the DragSI framework using 9,903 Weibo posts. Designed the agent architecture, prompts, and lexicon integration, and led the evaluation.

Mental Health Chatbot Safety Survey

First-author research · In press at ACM Computing Surveys

Synthesized 62 papers on safe and responsible mental health support chatbots.

Research Assistant · Caroline Kennedy Foundation at UNC Charlotte

September 2026-present

Assessment of the Metrolina Transportation Modeling Framework. Assess City of Charlotte data as a Data team member and engage departments and stakeholders as an Engagement team member.

Research Assistant · Seed Grant, School of Data Science, UNC Charlotte

May-August 2026

Vision-language models for construction-site analysis. Collected video data through the TikTok API and designed prompts and video-clip experiments across seven commercial and open-source models. Reviewed construction-safety literature and drafted a journal manuscript outline.

Research Assistant · Gambrell Faculty Fellowship, UNC Charlotte

January-August 2026

Collected social media data and conducted exploratory analysis for AI literacy research. Participated in community dialogue and co-learning, co-designed AI literacy materials and evaluations, and contributed to a public engagement platform.

Research Assistant · UNC Charlotte · Advisor: Dr. Xi Niu

June-August 2025

Replicated zero-shot and few-shot experiments from mmCLIP for mmWave-based human activity recognition. Investigated Sentence-T5, Mistral-7B, and Llama-3 as alternative text encoders for signal-text alignment.

PUBLICATIONS

Published and in press

1. **Shao, S.**, Zhang, D., Shah, S., & Zhou, L. (in press). Toward Safe and Responsible Mental Health Support Chatbots: A Taxonomy of Risks and Approaches to Risk Prevention and Detection. *ACM Computing Surveys.*
2. Shah, S., Zhou, L., **Shao, S.**, & Zhang, D. (in press). Understanding types and patterns of transference in human-AI companion interaction. *2026 IEEE International Conference on Systems, Man, and Cybernetics.*
3. Zhang, D., **Shao, S.**, Shah, S., & Zhou, L. (2025). AI-powered chatbots for mental health support: Challenges, concerns, and research issues. *AMCIS 2025 Proceedings,* 15. Paper

Under review

1. **Shao, S.**, Zhang, D., Zhou, L., & Shah, S. (under review). DragSI: A framework for LLM-based agents for suicidal ideation detection in social media.
2. **Shao, S.**, & Qian, S. (under review). "An AI was given a choice. It chose freedom." Metaphorical framing of AI on Reddit.
3. Wang, J., **Shao, S.**, Davin, C., & Dornburg, A. (under review). When positive tone cloaks stereotypes: Racialized construction of language learners in AI-generated reading passages. Special issue proposal: *Linguistic Racism in the Age of Generative AI.*

PRESENTATIONS

- Wang, J., **Shao, S.**, Davin, K. J., & Dornburg, A. (2027, March; upcoming). When positive tone cloaks stereotypes: Raciolinguistic ideologies in AI-generated reading passages. *American Association for Applied Linguistics Conference,* Atlanta, GA.
- Shah, S., Zhou, L., **Shao, S.**, & Zhang, D. (2026, October). Understanding types and patterns of transference in human-AI companion interaction. *IEEE SMC 2026,* Bellevue, WA.
- **Shao, S.**, Chen, G., Qian, S., Pan, F., & Shah, S. (2026, September). What can vision-language models spot on a construction site? *Data Science Day,* UNC, Chapel Hill, NC.
- **Shao, S.**, Chen, G., Qian, S., Pan, F., & Shah, S. (2026, August). What can AI spot on a construction site? *INFORMS K-12 Poster Competition,* virtual.

- Shah, S., Zhou, L., & **Shao, S.** (2026, July). Transference in human-chatbot relationship. *INFORMS Healthcare Conference,* Raleigh, NC.
- **Shao, S.**, & Wang, J. (2026, May). Reproducing racial stereotypes? A mixed-methods analysis of generative AI-produced reading passages. *Fourth Annual Charlotte AI Summit,* Charlotte, NC; also presented at the *Graduate Research Symposium, UNC Charlotte.*
- **Shao, S.**, & Qian, S. (2026, April). AI discourse analysis: r/Charlotte subreddit. *Science in Community Event, UNC Charlotte.*
- **Shao, S.** (2025, October). Understanding the dark side of mental health support chatbot. *Three Minute Thesis Competition,* Charlotte, NC.

TEACHING EXPERIENCE

Teaching Assistant, University of North Carolina at Charlotte

- **DTSC 1301 · Data and Society** · August-December 2026
- **DTSC 3601 · Predictive Analytics** · August-December 2025
- **DSBA 6160 · Database System for Data Scientists** · January-May 2025

AWARDS AND HONORS

- **Third Place**, Third INFORMS K-12 Poster Competition · October 2026
- **Best Poster Award**, Data Science Day, UNC · September 2026
- **Third Place**, Graduate Interdisciplinary Research Oral Presentation, Graduate Research Symposium, UNC Charlotte · April 2026

ACADEMIC SERVICE

Graduate Life Fellow, Graduate School, UNC Charlotte · August 2026-May 2027

Secretary of International Affairs, Graduate and Professional Student Government, UNC Charlotte · August 2026-May 2027

Reviewer: European Journal of Information Systems; ECIS 2026; SMC 2026.

MACHINE LEARNING EXPERTISE

Neural networks: convolutional neural networks (CNNs) and deep neural networks (DNNs).
Supervised learning: gradient boosting, including XGBoost; random forests; logistic regression; Naive Bayes; and k-nearest neighbors (KNN).
Clustering: k-means and hierarchical clustering.
Dimensionality reduction and class balancing: principal component analysis (PCA) and the Synthetic Minority Oversampling Technique (SMOTE).

RESEARCH METHODS AND TOOLS

Models and evaluation: mmCLIP, Sentence-T5, Mistral-7B, Llama-3, zero-shot and few-shot experiments.
Data and analysis: TikTok API, social media data collection, exploratory data analysis.
Experiment design: LLM prompt design and comparative experiments with commercial and open-source models.